WEEK 6 FINAL PROJECT REPORT

Integrated Data Science Project — Wine Class Classification

1. Executive summary

This project investigates whether chemical measurements can distinguish three wine classes in the public Wine Recognition dataset. The workflow integrates data validation, exploratory analysis, statistical hypothesis testing, machine-learning model development, evaluation and visual communication.

2. Problem statement

Given chemical measurements for a wine sample, predict which of the three known wine classes it belongs to. This is a supervised multiclass classification problem.

3. Dataset and source

The Wine Recognition dataset is available through scikit-learn and associated with the UCI Machine Learning Repository. It contains 178 observations, 13 numerical predictor features and three target classes. The code loads the dataset directly through load_wine(as_frame=True), avoiding a dependency on a manually located CSV file.

4. Data preparation and validation

The original target codes were mapped to descriptive labels class_0, class_1 and class_2. The data were checked for dimensions, missing values and duplicate rows. A stratified 75:25 train/test split with fixed random_state=42 was used for model evaluation. Scaling was included inside the Logistic Regression pipeline to avoid fitting the scaler on test data.

5. Exploratory data analysis

Class counts, numerical feature histograms and a correlation heatmap were generated. These visualizations describe class representation, feature distributions and linear relationships among chemical measurements.

6. Statistical analysis

A one-way ANOVA compared alcohol means across the three wine classes. The test produced $F = 135.0776$ and $p = 3.3195e\text{-}36$. Since $p < 0.05$, the null hypothesis of equal population means was rejected. ANOVA does not identify which class pairs differ; a post-hoc test would be needed.

7. Machine-learning modeling

Logistic Regression with StandardScaler and Random Forest were compared using the same stratified train/test split. Random Forest was then tuned using GridSearchCV and five-fold StratifiedKFold on the training data. Accuracy was used as the search scoring metric.

8. Evaluation results

Tuned Random Forest held-out test accuracy: 1.0000; macro precision: 1.0000; macro recall: 1.0000; macro F1: 1.0000. Best training cross-validation accuracy: 0.9852. Logistic Regression held-out accuracy on the same split: 1.0000.

9. Visual communication

The project includes a class distribution chart, alcohol distribution comparison, feature correlation heatmap, color-intensity boxplot and model confusion matrix. Each chart answers a specific question and supports interpretation of the dataset or model behavior.

10. Limitations and future work

The dataset is small and curated, so results may not generalize to external wine samples. ANOVA assumptions should be checked, and post-hoc tests could identify class-pair differences. Future work could use external validation, compare additional algorithms, calibrate predicted probabilities, analyze errors, and investigate feature selection.

11. Conclusion

This project demonstrates a complete and reproducible data-science workflow, from loading and validating a public dataset through EDA, statistical testing, model tuning, evaluation and visualization. The findings suggest that the chemical measurements contain information useful for distinguishing the three benchmark wine classes, while external validation is necessary before real-world deployment claims can be made.

12. References

UCI Machine Learning Repository — Wine dataset (https://archive.ics.uci.edu/dataset/109/wine).

Scikit-learn documentation — load_wine, train_test_split, Pipeline, GridSearchCV, RandomForestClassifier, LogisticRegression and metrics.

Pandas, Matplotlib and Seaborn documentation.